

# Manoj M Kadaramandalgi

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GenAI Engineer with 3+ years of experience at TCS GenAI CoE RnD Labs, building production-grade LLM applications, AI agents, workflow automation systems, and backend microservices.

Skilled in designing scalable FastAPI services, RAG pipelines, vector search, embeddings, and cloud-driven AI architectures.

Passionate about building intelligent systems that combine LLM orchestration, tool-calling, and automation to solve real business problems.

## WORK EXPERIENCE

### GenAI Developer

Aug 2022 - Present

#### Tata Consultancy Services (TCS)

India

- 3+ years of experience in Generative AI solution development at TCS GenAI Labs CoE.
- Developed a GenAI workflow builder that dynamically creates and deploys AI agents as modular Python files using FastAPI and JSON configurations without requiring application restart.
- Designed a plug-and-play agent ecosystem supporting RAG, embeddings, vector search, and LangChain workflows, enabling multi-step orchestration and automation.
- Reduced agent development time from hours to under 5 minutes by introducing dynamic imports, config-driven module loading, and runtime validation.
- Built scalable, system-design-driven architecture with centralized configuration, dynamic routing, observability, and seamless tool/model integration.

## SKILLS

### Programming Languages:

AI & GenAI: LangChain, RAG, Vector DBs, Embeddings, Prompt Engineering

Backend & System Design: FastAPI, REST APIs, Microservices

Languages: Python

Databases: PostgreSQL, SQLite

Frontend: nodejs, React

Workflow Automation: n8n, LangChain

Web Technologies: Fastapi, HTML, JavaScript

Cloud & DevOps: Azure Devops, Git

## CERTIFICATIONS

Hands-on Approach to AI, Cohort-2 by IIT Kharagpur AI4ICPS I HUB FOUNDATION

Oct 2024

Python Programming by Udemy

Completed internal training on Python, Generative AI, Docker.

• Self-learned LangChain, OpenAI integration, and vector DB operations through POCs and independent projects.

## EDUCATION

Bachelor of Engineering - Mechanical Engineering

Aug 2018 - Jun 2022

SDM College of Engineering and Technology, Dharwad, Karnataka

GPA: 8.42/10

## PROJECTS

### AI Agent Workflow System | FastAPI + LangChain + Python

- Built a dynamic agent creation platform where each agent is stored as a separate Python module and loaded at runtime.
- Added tool-calling, metadata routing, config-based agent definitions, and multi-step workflows.
- Supports RAG, document loaders, vector search, and custom tools.

### RAG Search & Question Answering Engine

- End-to-end retrieval pipeline using chunking, embeddings, vector DB (Chroma), and LLM prompt templates.
- Supports PDF/text ingestion, ranking, relevance scoring, and contextual answers.